VU Medical Zone

Amaan Khan

BT 603 Fermentation Technology

LECTURE 38 TO 73

FINAL TERM MERGE PPT

Media for Industrial Fermentations:

INHIBITORS

When certain inhibitors are added to fermentations, more of a specific product may be produced, or a metabolic intermediate which is normally metabolized is accumulated. One of the earliest examples is the microbial production of glycerol. Glycerol production depends on modifying the ethanol fermentation by removing acetaldehyde. The addition of sodium bisulphite to the broth leads to the formation of the acetaldehyde bisulphite addition compound (sodium hydroxy ethyl sulphite). Since acetaldehyde is no longer available for re-oxidation of $NADH_2$, its place as hydrogen acceptor is taken by dihydroacetone phosphate, produced during glycolysis. The product of this reaction is glycerol-3-phosphate, which is converted to glycerol.

Media for Industrial Fermentations

The application of general and specific inhibitors are illustrated in Table (in next slide). In most cases the inhibitor is effective in increasing the yield of the desired product and reducing the yield of undesirable related products.

A number of studies have been made with potential chlorination inhibitors, e.g. bromide, to minimize chlortetracycline production during a tetracycline fermentation.

Media for Industrial Fermentations

Specific and General Inhibitors used in fermentations

Product	Inhibitor	Main effect	Micro-organism
Glycerol	Sodium bisulphate	Acetaldehyde production repressed	*Saccharomyces cerivisiae*
Tetracycline	Bromide	Chlortetracycline formation repressed	*Streptomyces aureofaciens*
Glutamic acid	Penicillin	Dell wall permeability	*Micrococcus glutamicus*
Citric acid	Alkali metal/ phosphate, pH below 2.0	Oxalic acid repressed	*Aspergillus niger*
Valine	Various inhibitors	Various effects with different inhibitors	*Brevibacterium roseum*
Rifamycin B	Di-ethyl barbiturate	Other rifamycins inhibited	*Nocardia mediterranel*

Media for Industrial Fermentations

Inhibitors have also been used to affect cell-wall structure and increase the permeability for release of metabolites. The best example is the use of penicllin and surfactants in glutamic acid production.

END

Advances in Fermentation Technology

Media for Industrial Fermentations:

INDUCERS-1

Media for Industrial Fermentations

The majority of enzymes which are of industrial interest are inducible. Induced enzymes are synthesized only in response to the presence in the environment of an inducer. Inducers are often substrates such as starch or dextrins for amylases, maltose for pullulanase and pectin for pectinases.

Some inducers are very potent, such as isovaleronitrile inducing nitralase. Substrate analogues that are not attacked by the enzyme may also serve as enzyme inducers. Examples of industrially important enzyme inducers are shown in Table given in next slide.

Media for Industrial Fermentations

Some examples of industrially important enzyme inducers

Enzyme	Inducer	Micro-organism
α-Amylase	Starch Maltose	*Aspergillus spp.* *Bacillus subtilis*
Pullulanase	Maltose	*Aerobacter aerogenes*
α-Mannosidase	Yeast mannans	*Streptomyces griseus*
Penicillin acylase	Phenylacetic acid	*Escherichia coli*
Proteases	Various proteins	*Bacillus spp.* *Streptococcus spp.* *Streptomyces spp.* *Aspergillus spp.*
Cellulase	Cellulose	*Trichoderma viridae*
Pectinases	Pectin (beet pulp, apple pomace, citrus peel)	*Aspergillus spp.*
Nitralase	Isovaleronitrile	*Rhodococcus rhodochrous*

Media for Industrial Fermentations

One unusual application of an inducer is the use of yeast mannan in streptomycin production. During the fermentation varying amounts of streptomycin and mannosidostreptomycin are produced.

Since mannosidostreptomycin has only 20% of the biological activity of streptomycin, the former is an undesirable product.

The production organism *Streptomyces griseus* can be induced by yeast mannan to produce *β*-mannosidase which will convert mannosidostreptomycin to streptomycin.

Media for Industrial Fermentations

Most inducers which are included in microbial enzyme media are substrates or substrate analogues, but intermediates and products may sometimes be used as inducers. For example, maltodextrins will induce amylase and fatty acids induce lipase. However, the cost may prohibit their use as inducers in a commercial process.

END

Media for Industrial Fermentations

The types of antifoams mentioned in previous slides are generally added when foaming occurs during the fermentation. Because many antifoams are of low solubility they need a carrier such as lard oil, liquid paraffin or castor oil, which may be metabolized and affect the fermentation process.

END

Media for Industrial Fermentations:
INDUCERS-2

It is now possible to produce a number of heterologous proteins in yeasts, fungi and bacteria. These include proteins of viral, human, animal, plant and microbial origin. However, heterologous proteins may show some degree of toxicity to the host and have a major influence on the stability of heterologous protein expression. As well as restricting cell growth as biomass the toxicity will provide selective conditions for segregant cells which no longer synthesize the protein at such a high level.

Therefore, optimum growth conditions may be achieved by not synthesizing a heterologous protein continuously and only inducing it after the host culture has grown up in a vessel to produce sufficient biomass. In cells of *Saccharomyces cerevisiae* where the *Gal*1 promoter is part of the gene expression system, product formation may be induced by galactose addition to the growth medium which contains glycerol or low non-repressing levels of glucose as a carbon source.

Methylotrophic yeasts such as *Hansenula polymorpha* and *Pichia pastoris* may be used as alternative systems because of the presence of an alcohol oxidase promoter.

During growth on methanol, which also acts as an inducer, the promoter is induced to produce about 30% of the cell protein. In the presence of glucose or ethanol it is undetectable.

END

Media for Industrial Fermentations:

OXYGEN REQUIREMENTS, TRACE ELEMENTS, pH and OSMOLALITY

It is sometimes forgotten that oxygen, although not added to an initial medium as such, is nevertheless a very important component of the medium in many processes, and its availability can be extremely important in controlling growth rate and metabolite production.

1. ***Fast metabolism.*** ***The culture may become oxygen*** limited because sufficient oxygen cannot be made available in the fermenter if certain substrates, such as rapidly metabolized sugars which lead to a high oxygen demand, are available in high concentrations.
2. ***Rheology.*** The individual components of the medium can influence the viscosity of the final medium and its subsequent behaviour with respect to aeration and agitation.
3. ***Antifoams.*** Many of the antifoams in use will as surface active agents and reduce the transfer rate.

❑ *Trace Elements*

The role of trace elements in medium formulation can be significant. Cultured cells normally require Fe, Zn, Cu, Se, Mn, Mo and V. These are often present as impurities in other media components. Cleveland *et al.* (1983) found that if the number of trace elements were increased, insulin, transferrin, albumin and liposomes were not needed in a serum-free hybridoma medium. They included Al, Ag, Ba, Br, Cd, Co, Cr, F, Ge, J, Rb, Zr, Si, Ni and Sn as well as those previously mentioned.

Media for Industrial Fermentations

❑ *pH*

The normal buffer system in tissue culture media is the CO_2-bicarbonate system. This is a weak buffering system and can be improved by the use of a zwitterinic buffer such as Hepes, either in addition to or instead of the CO_2-bicarbonate buffer. Continuous pH control is achieved by the addition of sodium bicarbonate or sodium hydroxide (with fast mixing) when too acid. The pH does not normally become too Alkaline so acid additions are not required but provision may be made for CO_2 additions.

Media for Industrial Fermentations

❑ *Osmolality*

The optimum range of osmotic pressure for growth is often quite narrow and varies with the type of cell and the species from which it was isolated. It maybe necessary to adjust the concentration of NaCl when major additions are made to a medium.

END

Advances in Fermentation Technology

Media for Industrial Fermentations:

ANTIFOAMS-1

Media for Industrial Fermentations

- In most microbiological processes, foaming is a problem may be due to a component in the medium or some factor produced by the micro-organism.
- The most common cause of foaming is due to proteins in the medium, such as corn-steep liquor, meal, soybean meal, yeast extract or meat extract.
- These proteins may denature at the air-broth interface and form a skin which does not rupture readily.
- The foaming can cause removal of cells from the medium which will lead to autolysis and the further release of microbial cell proteins will probably increase the stability of the foam.

Media for Industrial Fermentations

- If uncontrolled, then numerous changes may occur and physical and biological problems may be created. These include reduction in the working volume of the fermenter due to oxygen exhausted gas bubbles circulating in the system changes in bubble size, lower mass and heat transfer rates, invalid process data due to interference at sensing electrodes and incorrect monitoring and control.
- The biological problems include deposition of cells in upper parts of the fermenter, problems of sterile operation with the air filter exits of the fermenter becoming wet, and there is danger of microbial infection and the possibility of siphoning leading to loss of product.

Media for Industrial Fermentations

Hall *et al.* (1973) have recognized five patterns of foaming in fermentations:

1. Foaming remains at a constant level throughout the fermentation. Initially it is due to the medium and later due to microbial activity.
2. A steady fall in foaming during the early part of the fermentation, after which it remains constant. Initially it is due to the medium but there are no later effects caused by the micro-organism.
3. The foaming falls slightly in the early stages of the fermentation then rises. There are very slight effects caused by the medium but the major effects are due to microbial activity.
4. The fermentation has a low initial foaming capacity which rises. These effects are due solely to microbial activity.

Media for Industrial Fermentations

5. A more complex foaming pattern during the fermentation which may be a combination of two or more of the previously described patterns.

END

Advances in Fermentation Technology

Media for Industrial Fermentations:

ANTIFOAMS-2

Media for Industrial Fermentations

If excessive foaming is encountered there are three ways of approaching the problem:

1. To try and avoid foam formation by using a defined medium and a modification of some of the physical parameters (pH, temperature, aeration and agitation). This assumes that the foam is due to a component in the medium and not a metabolite.

2. The foam is unavoidable and antifoam should be used. This is the more standard approach.

3. To use a mechanical foam breaker.

Media for Industrial Fermentations

Antifoams are surface active agents, reducing the surface tension in the foams and destabilizing protein films by

(a) hydrophobic bridges between two surfaces, (b) displacement of the absorbed protein, and (c) rapid spreading on the surface of the film.

Media for Industrial Fermentations

An ideal antifoam should have the following properties:

1. Should disperse readily and have fast action on an existing foam.
2. Should be active at low concentrations.
3. Should be long acting in preventing new foam formation.
4. Should not be metabolized by the microorganism.
5. Should be non-toxic to the micro-organism.
6. Should be non-toxic to humans and animals.
7. Should not cause any problems in the extraction and purification of the product.
8. Should not cause any handling hazards.
9. Should be cheap.

Media for Industrial Fermentations

10. Should have no effect on oxygen transfer.

11. Should be heat sterilizable.

END

Media for Industrial Fermentations:

ANTIFOAMS-3

The following compounds which meet most of ideal antifoam requirements (as discussed in previous module) have been found to be most suitable in different fermentation processes:

1. Alcohols; stearyl and octyl decanol.
2. Esters.
3. Fatty acids and derivatives, particularly glycerides, which include cottonseed oil, linseed oil, say-bean oil, olive oil, castor oil, sunflower oil, rapeseed oil and cod liver oil.
4. Silicones.
5. Sulphonates.
6. Miscellaneous; Alkaterge C, oxazaline, poly-propylene glycol.

Unfortunately, the concentrations of many antifoams which are necessary to control fermentations will reduce the oxygen-transfer rate by as much as 50%; therefore antifoam additions must be kept to an absolute minimum. There are also other antifoams which will increase the oxygen-transfer rate. If the oxygen-transfer rate is severely affected by antifoam addition then mechanical foam breakers may have to be considered as a possible alternative. Various studies concluded that foam control in industry is still an empirical art. The best method for a particular process in one factory is not necessarily the best for the same process on another site. The design and operating parameters of a fermenter may affect the properties and quantity of foam formed.

The types of antifoams mentioned in previous slides are generally added when foaming occurs during the fermentation. Because many antifoams are of low solubility they need a carrier such as lard oil, liquid paraffin or castor oil, which may be metabolized and affect the fermentation process.

END

Media for Industrial Fermentations: MEDIUM OPTIMIZATION

When considering the biomass growth phase in isolation it must be recognized that efficiently grown biomass produced by an 'optimized' high productivity growth phase is not necessarily best suited for its ultimate purpose, such as synthesizing the desired product.

Different combinations and sequences of process conditions need to be investigated to determine the growth conditions which produce the biomass with the physiological state best constituted for product formation. There may be a sequence of phases each with a specific set of optimal conditions.

Medium optimization by the classical method of changing one independent variable (nutrient, antifoam, pH, temperature, etc.) while fixing all the others at a certain level can be extremely time consuming and expensive for a large number of variables.

To make a full factorial search which would examine each possible combination of independent variable at appropriate levels could require a large number of x^n experiments, where

x is the number of levels and

n is the number of variables.

This may be quite appropriate for three nutrients at two concentrations (2^3 =8 trials) but not for six nutrients at three concentrations (3^6 =729)

Industrially the aim is to perform the minimum number of experiments to determine optimal conditions.

Media for Industrial Fermentations

Other alternative strategies can therefore be considered which allow more than one variable to be changed at a time.

END

Advances in Fermentation Technology

Media for Industrial Fermentations:

MEDIUM OPTIMIZATION-

Plackett-Burman Design-1

Media for Industrial Fermentations

When more than five independent variables are to be investigated, the Plackett-Burman design may be used to find the most important variables in a system, which are then optimized in further studies (Plackett and Burman, 1946). These authors give a series of designs for up to one hundred experiments using an experimental rationale known as balanced incomplete blocks. This technique allows for the evaluation of X-1 variables by X experiments. X must be a multiple of 4, e.g. 8, 12, 16, 20, 24, etc.

Media for Industrial Fermentations

Normally one determines how many experimental variables need to be included in an investigation and then selects the Plackett-Burman design which meets that requirement most closely in multiples of 4.

Any factors not assigned to a variable can be designated as a dummy variable.

Alternatively, factors known to not have any effect may be included and designated as dummy variables.

Table in the next slide shows a Plackett-Burman design for seven variables *(A -G)* at high and low levels in which two factors, *E* and *G*, are designated as 'dummy' variables.

Media for Industrial Fermentations

TABLE *Plackett–Burman design for seven variables*

Trial	Variables							Yield
	A	*B*	*C*	*D*	*E*	*F*	*G*	
1	H	H	H	L	H	L	H	1.1
2	L	H	H	H	L	H	L	6.3
3	L	L	H	H	H	L	H	1.2
4	H	L	L	H	H	H	L	0.8
5	L	H	L	L	H	H	H	6.0
6	H	L	H	L	L	H	H	0.9
7	H	H	L	H	L	L	H	1.1
8	L	L	L	L	L	L	L	1.4

H denotes a high level value; L denotes a low level value.

Each horizontal row represents a trial and each vertical column represents the H (high) and L (low) values of one variable in all the trials.

Media for Industrial Fermentations

Consider the variable *A*; for the trials in which *A* is high, *B* is high in two of the trials and low in the other two. Similarly, *C* will be high in two trials and low in two, as will all the remaining variables.

For those trials in which *A* is low, *B* will be high two times and low two times. This will also apply to all the other variables.

Thus, the effects of changing the other variables cancel out when determining the effect of *A*. The same logic then applies to each variable.

However, no changes are made to the high and low values for the *E* and *G* columns.

Media for Industrial Fermentations

The effects of the dummy variables are calculated in the same way as the effects of the experimental variables.

If there are no interactions and no errors in measuring the response, the effect shown by a dummy variable should be 0. If the effect is not equal to 0, it is assumed to be a measure of the lack of experimental precision plus any analytical error in measuring the response. This procedure will identify the important variables and allow them to be ranked in order of importance to decide which to investigate in a more detailed study to determine the optimum values to use.

Media for Industrial Fermentations

This procedure will identify the important variables and allow them to be ranked in order of importance to decide which to investigate in a more detailed study to determine the optimum values to use.

END

Media for Industrial Fermentations:

MEDIUM OPTIMIZATION-

Plackett-Burman Design-2

The stages in analyzing the data (mentioned in Table in previous module; and in the Table given in next slide) using Nelson's (1982) example are as follows:

1. Determine the difference between the average of the H (high) and L (low) responses for each independent and dummy variable. Therefore the difference =

$$\Sigma A\,(H) - \Sigma A\,(L)$$

TABLE *Plackett–Burman design for seven variables*

Trial	Variables							Yield
	A	*B*	*C*	*D*	*E*	*F*	*G*	
1	H	H	H	L	H	L	H	1.1
2	L	H	H	H	L	H	L	6.3
3	L	L	H	H	H	L	H	1.2
4	H	L	L	H	L	H	L	0.8
5	L	H	L	L	H	H	H	6.0
6	H	L	H	L	L	H	H	0.9
7	H	H	L	H	L	L	H	1.1
8	L	L	L	L	L	L	L	1.4

H denotes a high level value; L denotes a low level value.

Each horizontal row represents a trial and each vertical column represents the H (high) and L (low) values of one variable in all the trials.

Analysis of the Yields

	Factor						
	A	*B*	*C*	*D*	*E*	*F*	*G*
Σ(H)	3.9	14.5	9.5	9.4	9.1	14.0	9.2
Σ(L)	14.9	4.3	9.3	9.4	9.7	4.8	9.6
Difference	−11.0	10.2	0.2	0.0	−0.6	9.2	−0.4
Effect	−2.75	2.55	0.05	0.00	−0.15	2.30	−0.10
Mean square	15.125	13.005	0.005	0.000	0.045	10.580	0.020

$$\text{Mean square for 'error'} = \frac{0.045 + 0.020}{2} = 0.0325$$

Media for Industrial Fermentations

The effect of an independent variable on the response is the difference between the average response for the four experiments at the high level and the average value for four experiments at the low level. Thus the effect of

$$A = \frac{\Sigma A\,(H)}{4} - \frac{\Sigma A\,(L)}{4}$$

$$= \frac{2(\Sigma A\,(H) - \Sigma A\,(L))}{8}$$

This value should be near zero for the dummy variables.

Media for Industrial Fermentations

2. Estimate the mean square of each variable (the variance of effect). For A the mean square will be = $\dfrac{(\Sigma A\,(H) - \Sigma A\,(L))^2}{8}$

3. The experimental error can be calculated by averaging the mean squares of the dummy effects of *E* and *G*. Thus, the mean square for error = $\dfrac{0.045 + 0.020}{2} = 0.0325$

This experimental error is not significant.

Media for Industrial Fermentations

4. The final stage is to identify the factors which are showing large effects. In the example this was done using an *F*-test for

$$\frac{\text{Factor mean square.}}{\text{error mean square.}}$$

This gives the following values:

$$A = \frac{15.125}{0.0325} = 465.4, \qquad C = \frac{0.0500}{0.0325} = 1.538$$

$$B = \frac{13.005}{0.0325} = 400.2, \qquad D = \frac{0.0000}{0.0325} = 0.00$$

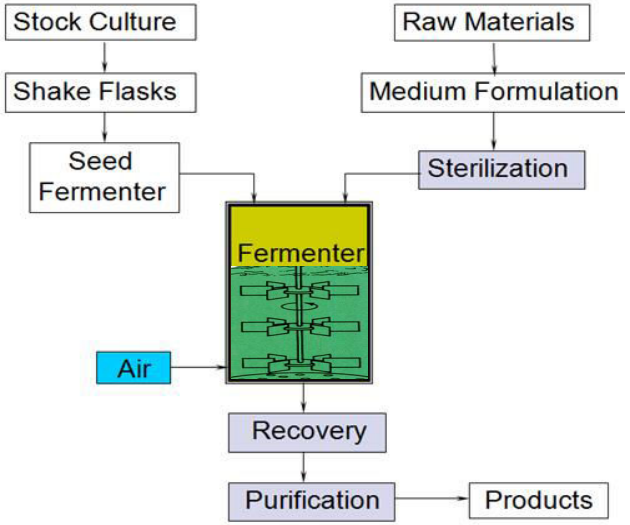
$$F = \frac{10.580}{0.0325} = 325.6.$$

Media for Industrial Fermentations

When Probability Tables are examined it is found that Factors *A*, *B* and *F* show large effects which are very significant, whereas *C* shows a very low effect which is not significant and *D* shows no effect. *A*, *B* and *F* have been identified as the most important factors.

END

Aeration & Agitation: Introduction-1



The majority of fermentation processes are aerobic and, therefore, require the provision of oxygen. If the stoichiometry of respiration is considered, then the oxidation of glucose may be represented as:

$$C_6H_{12}O_6 + 6O_2 = 6H_2O + 6CO_2$$

1 mole + 6mole

180g 192g

However, both components must be in solution before they are available to a micro-organism and oxygen is approximately 6000 times less soluble in water than is glucose.

A fermentation medium saturated with oxygen contains approximately

7.6 mg dm^{-3} of oxygen at 30°C

Thus, it is not possible to provide a microbial culture with all the oxygen it will need for the complete oxidation of the glucose (or any other carbon source) in one addition.

Therefore, a microbial culture must be supplied with oxygen during growth at a rate sufficient to satisfy the organisms' demand.

Aeration & Agitation

The oxygen demand of an industrial fermentation process is normally satisfied by aerating and agitating the fermentation broth.

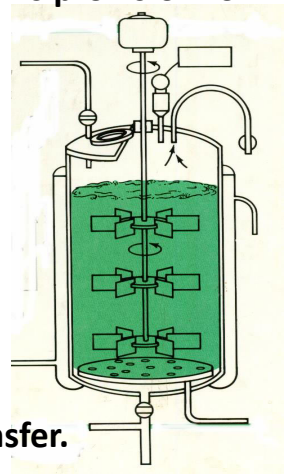
END

Advances in Fermentation Technology

Aeration & Agitation:
Introduction-2

Aeration & Agitation

Important factor in a fermenters is provision for adequate mixing of its contents.



Mixing in fermentation

- to disperse the air bubbles
- to suspend the cells
- to enhance heat and mass transfer in the medium

All relate to Gas-liquid mass transfer.

Aeration & Agitation

Aeration refers to the process of introducing air to increase oxygen concentration in liquids. Aeration may be performed by bubbling air through the liquid, spraying the liquid into the air or agitation of the liquid to increase surface absorption.

Supplying oxygen to aerobic cells is a significant challenge.
The problem is that oxygen is poorly soluble in water .
The solubility of oxygen in pure water is 8 mg/L at 4°C.
The solubility of oxygen decreases as with increasing temperature and concentration of solutes in the solution.

Aeration & Agitation

- Supply of oxygen is the rate limiting step in an aerobic fermentation
- Satisfy oxygen demands constitute a large proportion of the operating and capital cost of a industrial scale fermentation system.

END

Advances in Fermentation Technology

Aeration & Agitation:

Oxygen Requirements of Industrial Fermentations-1

Aeration & Agitation

Although a consideration of the stoichiometry of respiration gives an appreciation of the problem of oxygen supply, it gives no indication of an organism's true oxygen demand as it does not take into account the carbon that is converted into biomass and products.

A number of workers have considered the overall stoichiometry of the conversion of oxygen, a source of carbon and a source of nitrogen into biomass and have used such relationships to predict the oxygen demand of a fermentation.

Aeration & Agitation

Stoichiometric equations describing oxygen demand in a fermentation

$$6.67CH_2O + 2.1O_2 = C_{3.92}H_{6.5}O_1 + 2.75CO_2 + 3.42H_2O$$

$$7.14CH_2 + 6.135O_2 = C_{3.92}H_{6.5}O_1 + 3.22CO_2 + 3.89H_2O$$

Terms used

$C_{3.92}H_{6.5}O_{1.94}$ is 100 g (dry weight) of yeast cells; CH_2O is carbohydrate

CH_2 is hydrocarbon

Reference: Darlington (1964)

Aeration & Agitation

Stoichiometric equations describing oxygen demand in a fermentation (Cont.....)

$$C = (A/Y) - B$$

A = Amount of oxygen for combustion of 1 g of substrate to CO_2, H_2O and NH_3, if nitrogen is present in the substrate

B = Amount of oxygen required for the combustion of 1 g cells to CO_2, H_2O and NH_3

Y = Cell yield (g cells g^{-1} substrate)

C = g oxygen consumed for the production of 1 g of cells

Reference: Johnson (1964)

Aeration & Agitation

Stoichiometric equations describing oxygen demand in a fermentation (Cont.....)

$$(dx/dt)/Y + mx + p = (dCO_2/dt) = -(dO_2/dt)RQ$$

x = biomass concentration

t = time

Y = g biomass g^{-1} carbon substrate

m = maintenance

p = allowance for antibiotic production

Reference: Righelato et al., (1968)

Aeration & Agitation

Stoichiometric equations describing oxygen demand in a fermentation (Cont.....)

$$Y_{O/P} = (0.53/Y_{P/G}) - (0.6X/P) - 0.43$$

$Y_{O/P}$ = g oxygen consumed g^{-1} glucose

$Y_{P/G}$ = g sodium penicillin G produced g^{-1} glucose

X = g cells (dry weight) produced

P = g sodium penicillin G produced

Reference: Cooney (1979)

Aeration & Agitation

Stoichiometric equations describing oxygen demand in a fermentation (Cont.....)

$$Y_O = \{(32C + 8H - 16O)/YM\} - 1.58$$

Y_O = g oxygen consumed g^{-1} cells produced

Y = Cell yield (g cells g^{-1} substrate)

M = Molecular weight of the carbon source

C, H and O = Number of atoms of carbon, hydrogen and oxygen per molecule of carbon source

Reference: Mateles (1971)

Aeration & Agitation

From Darlington's and Johnson's equations (in above mentioned Table) it may be seen that the production of 100 grams of biomass from hydrocarbon requires approximately three times the amount of oxygen to produce the same amount of biomass from carbohydrate.

END

Advances in Fermentation Technology

Aeration & Agitation:

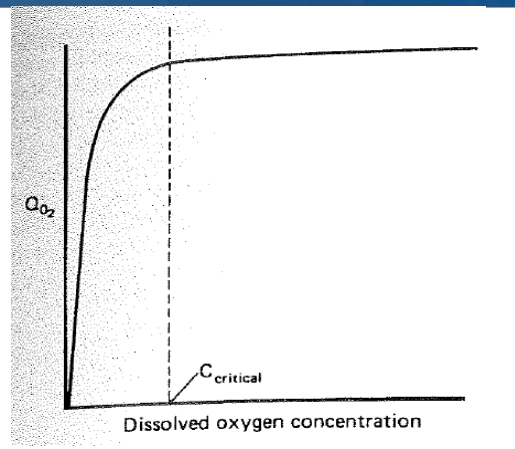
Oxygen Requirements of Industrial Fermentations-2

Aeration & Agitation

It must be remembered that the high carbon content of hydrocarbon substrates means that high yield factors (g biomass g^{-1} substrate consumed) are obtained and the decision to use such substrates is based on the balance between the advantage of high biomass yield and the disadvantage of high oxygen demand and heat generation.

However, it is inadequate to base the provision of oxygen for a fermentation simply on an estimation of overall demand, because the metabolism of the culture is affected by the concentration of dissolved oxygen in the broth. The effect of dissolved oxygen concentration on the specific oxygen uptake rate (Q_{O_2} mmoles of oxygen consumed per gram dry weight of cells per hour) has been shown to be of the Michaelis-Menten type, as shown in Fig. (in next slide).

Aeration & Agitation



The effect of dissolved oxygen concentration on the Q_{O_2} of a microorganism

It may be seen that the specific oxygen uptake rate increases with increase in the dissolved oxygen concentration up to a certain point (referred to as $C_{critical}$) above which no further increase in oxygen uptake rate occurs.

Aeration & Agitation

Critical dissolved oxygen concentrations for a range of microorganisms

Organism	Temperature	Critical dissolved oxygen concentration (mmoles dm^{-3})
Azotobacter sp.	30	0.018
Escherichia coli	37	0.008
Saccharomyces sp.	30	0.004
Penicillium chrysogenum	24	0.022

Aeration & Agitation

Oxygen requirements of a range of microorganisms grown on a range of substrates

Substrate	Organism	Oxygen requirement (g O_2 g^{-1} dry wt.)
Glucose	*Escherichia coli*	0.4
Methanol	*Pseudomonas C*	1.2
Octane	*Pseudomonas* sp.	1.7

Aeration & Agitation

Thus, maximum biomass production may be achieved by satisfying the organism's maximum specific oxygen demand by maintaining the dissolved oxygen concentration greater than the critical level. *If the dissolved oxygen concentration were to fall below the critical level then the cells may be metabolically disturbed.*

END

Advances in Fermentation Technology

Aeration & Agitation:
Henry's Law and Oxygen Supply

Aeration & Agitation

Henry's Law describes the solubility of O_2 in nutrient solution in relation to the O_2 partial pressure in the gas phase

$$C^* = \frac{P_o}{H}$$

C^* is the oxygen saturation concentration of the nutrient solution,

P_o is the partial pressure of the gas in the gas phase

H is Henry's constant, which is specific for the gas and the liquid phase.

Aeration with air 9 mg O_2/L dissolves in water, with pure oxygen 43 mg O_2/L.

Aeration & Agitation

❑Oxygen Supply

Oxygen is normally supplied to microbial cultures in the form of air, this being the cheapest available source of the gas. The method for provision of a culture with a supply of air varies with the scale of the process:

- Laboratory-scale cultures may be aerated by means of the shake-flask technique where the culture (50 to 100 cm^3) is grown in a conical flask (250 to 500 cm^3) shaken on a platform contained in a controlled environment chamber.
- Pilot- and industrial-scale fermentations are normally carried out in stirred, aerated vessels (fermenters).

Aeration & Agitation

Bartholomew *et al*. (1950) represented the transfer of oxygen from air to the cell, during a fermentation, as occurring in a number of steps:

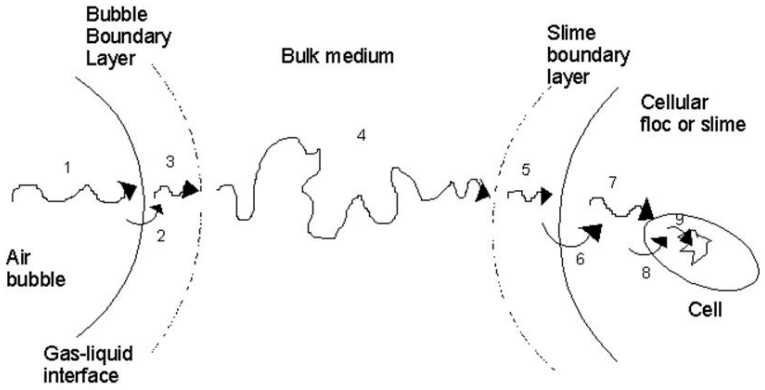
i. The transfer of oxygen from an air bubble into solution.
ii. The transfer of the dissolved oxygen through the fermentation medium to the microbial cell.
iii. The uptake of the dissolved oxygen by the cell.

These findings have been shown to be correct for non-viscous fermentations but it has been demonstrated that transfer may be limited by some other stages in certain highly viscous fermentations.

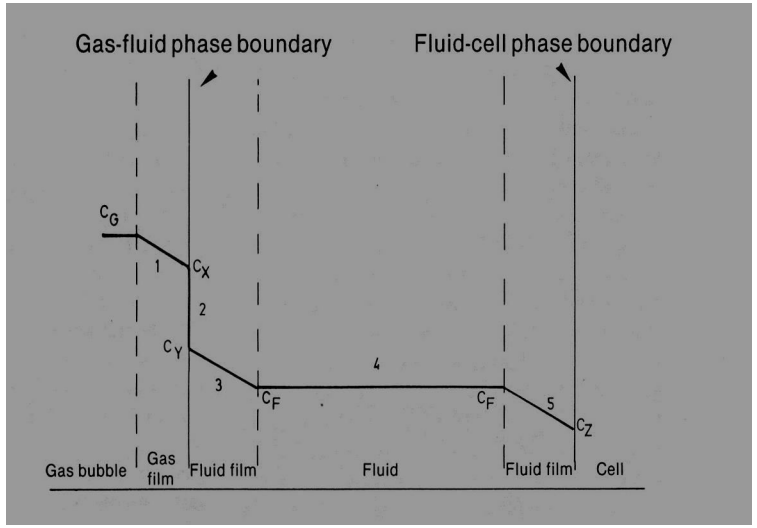
Aeration & Agitation

Oxygen Path

(From A Bubble To An Immobilized Cell System)



Aeration & Agitation



Aeration & Agitation

The requirement for a high dissolved oxygen concentration by many fermentations has resulted in the development of process techniques to ensure that the fermentation does not exceed the oxygen-supply capabilities of the fermentation vessel.

END

Advances in Fermentation Technology

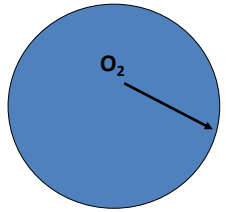
Aeration & Agitation:

Oxygen Transfer Process-1

Aeration & Agitation

Step 1

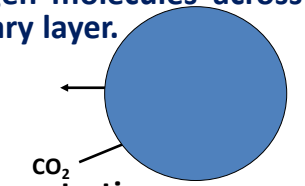
Diffusion through the bubble to the gas liquid interface. Gas molecules move quickly, they are evenly distributed throughout the bubble.



Step 2

Diffusion across the gas-liquid interface. This step will be very rapid if the concentration of oxygen in the bubble high. High oxygen concentrations in the bubble (as measured in terms of partial pressure) will push the oxygen molecules across the interface, into the boundary layer.

If the medium is rich in CO_2, then the carbon dioxide will be pushed into the bubble.



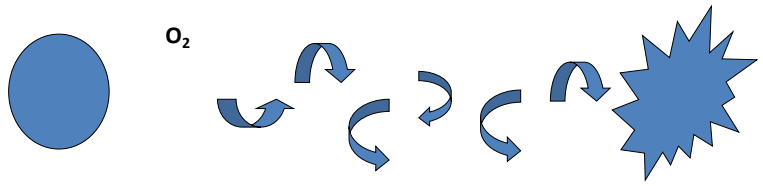
The bubble contains a low concentration of oxygen, then the rate of oxygen transfer out of the bubble will be slow or even zero.

Step 3

Diffusion through the bubble boundary layer. The movement of solutes through the boundary layer is slow. Solutes move through the liquid by diffusion. The movement of the molecule will be driven by the concentration gradient across the boundary layer. Factors affect the rate of diffusion of oxygen through the boundary layer, including: temperature, concentration of oxygen in the bulk liquid, saturation concentration of oxygen in the liquid, concentration of oxygen in the bubble, size of the molecule and viscosity of the medium.

Step 4

Movement through the bulk liquid by forced convection and diffusion. The rate of movement of an oxygen molecule through the bulk liquid is dependent on the degree of mixing (relative to the volume of the reactor) and viscosity of the medium.



Step 5-9: Movement through the floc

- Step 5 Movement through the boundary layer surrounding the microbial slime
- Step 6 Entry into the slime
- Step 7 Movement through the slime
- Step 8 Movement across the cell membrane
- Step 9 Reaction

Step 5 and Step 7 are slow processes.

If only, suspended cells are involved the level of mixing in the bulk liquid is sufficiently high;

Then, the rate limiting step in the oxygen transfer process is the movement of the oxygen molecules through the bubble boundary layer.

END

Aeration & Agitation:

Oxygen Transfer Process-2

The rate of oxygen transfer from air bubble to the liquid phase may be described by the equation:

$$dC_L/dt = K_L a(C^* - C_L)$$

where C_L is the concentration of dissolved oxygen in the fermentation broth (mmoles dm^{-3}),

t is time (hours).

dC_L/dt is the change in oxygen concentration over a time period, i.e. the oxygen-transfer rate (mmoles O_2 dm^{-3} h^{-1}),

K_L is the mass transfer coefficient (cm h^{-1}),

a is the gas/liquid interface area per liquid volume (cm^2 cm^{-3}),

C^* is the saturated dissolved oxygen concentration (mmoles dm^{-3}).

It is extremely difficult to measure both 'K_L' and 'a' in a fermentation and, therefore, the two terms are generally combined in the term $K_L a$, known as the volumetric mass-transfer coefficient. The units of $K_L a$, are reciprocal time (h^{-1}).

The volumetric mass-transfer coefficient is used as a measure of the aeration capacity of a fermenter. The larger the $K_L a$, the higher the aeration capacity of the system.

The $K_L a$ value will depend upon the design and operating conditions of the fermenter and will be affected by the variables such as aeration rate, agitation rate and impeller design.

Aeration & Agitation

Oxygen transfer rate (OTR) is the critical criterion in a fermentation and, as may be seen from equation in previous slide, it is affected by both K_La and dissolved oxygen concentration.

The dissolved oxygen concentration reflects the balance between the supply of dissolved oxygen by the fermenter and the oxygen demand of the organism. If the K_La of the fermenter is such that the oxygen demand of the organism cannot be met, the dissolved oxygen concentration will decrease below the critical level (C_{crit}).

If the K_La is such that the oxygen demand of the organism can be easily met the dissolved oxygen concentration will be greater than C_{crit} and may be as high as 70 to 80% of the saturation level.

Aeration & Agitation

Thus, the K_La of the fermenter must be such that the optimum oxygen concentration for product formation can be maintained in solution throughout the fermentation.

END

Advances in Fermentation Technology

Aeration & Agitation:

Determination of K_La values-

Sulphite Oxidation Technique

Aeration & Agitation

Determination of the K_La of a fermenter is essential in order to establish its aeration efficiency and to quantify the effects of operating variables on the provision of oxygen. It is important to remember at this stage that dissolved oxygen is usually monitored using a dissolved oxygen electrode which records dissolved oxygen activity or dissolved oxygen tension (DOT) whilst the equations describing oxygen transfer are based on dissolved oxygen concentration.

The solubility of oxygen is affected by dissolved solutes so that pure water and a fermentation medium saturated with oxygen would have different dissolved oxygen concentrations yet have the same DOT, i.e. an oxygen electrode would record 100% for both. Thus, to translate DOT into concentration the solubility of oxygen in the fermentation medium must be known and this can present difficulties.

❑ Sulphite Oxidation Technique

This technique does not require the measurement of dissolved oxygen concentrations but relies on the rate of conversion of a 0.5 M solution of sodium sulphite to sodium sulphate in the presence of a copper or cobalt catalyst:

$$Na_2SO_3 + 0.5O_2 = Na_2SO_4$$

As oxygen enters solution it is immediately consumed in the oxidation of sulphite, so that the sulphite oxidation rate is equivalent to the oxygen-transfer rate. Since the dissolved oxygen concentration, is zero then the K_La may then be calculated from the equation:

$$OTR = K_La \cdot C^*$$

(where OTR is the oxygen transfer rate)

The volumes of the thiosulphate titrations are plotted against sample time and the oxygen transfer rate may be calculated from the slope of the graph.

END

Aeration & Agitation:

Determination of K_La values-

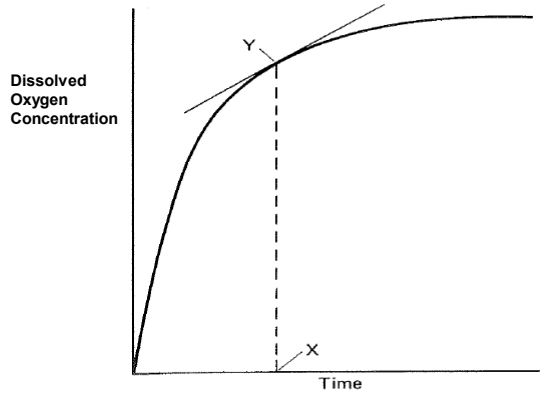
Gassing-out Techniques-1

❑ Gassing-out Techniques

The estimation of the K_La of a fermentation system by gassing-out techniques depends upon monitoring the increase in dissolved oxygen concentration of a solution during aeration and agitation.

The oxygen transfer rate will decrease during the period of aeration as C_L approaches C^* due to the decline in the driving force $(C^*–C_L)$. The oxygen transfer rate, at anyone time, will be equal to the slope of the tangent to the curve of values of dissolved oxygen concentration against time of aeration, as shown in Figure on next slide.

Aeration & Agitation



The increase in dissolved oxygen concentration of a solution over a period of aeration. The oxygen transfer rate at time X is equal to the slope of the tangent at point Y.

Aeration & Agitation

To monitor the increase in dissolved oxygen over an adequate range it is necessary first to decrease the oxygen level to a low value. Two methods have been employed to achieve this lowering of the dissolved oxygen concentration - the static method and the dynamic method.

Aeration & Agitation

i. Static Method of Gassing-out

In this technique, the oxygen concentration of the solution is lowered by gassing the liquid out with nitrogen gas, so that the solution is 'scrubbed' free of oxygen. The deoxygenated liquid is then aerated and agitated and the increase in dissolved oxygen monitored using some form of dissolved oxygen probe. The increase in dissolved oxygen concentration has already been described by equation:

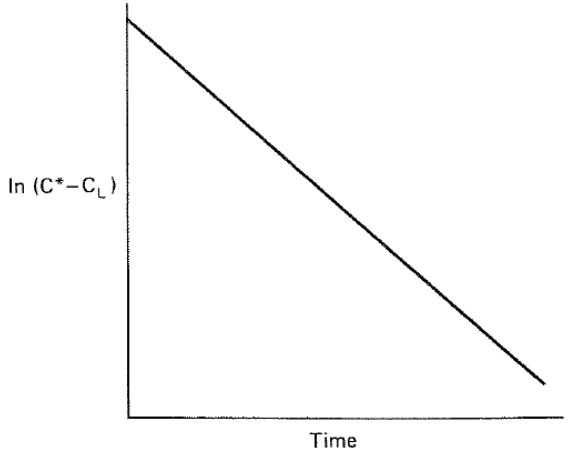
$$dC_L/dt = K_L a(C^* - C_L)$$

Integration of above equation yields:

$$\ln(C^* - C_L) = -K_L at.$$

Aeration & Agitation

Thus, a plot of ln $(C^* - C_L)$ against time will yield a straight line of slope $K_L a$, as shown in following figure:



This method is acceptable for small scale vessels, there are severe limitations to its use on large scale fermenters which have high gas residence times.

When the air supply to such a vessel is resumed after deoxygenation with nitrogen, the oxygen concentration in the gas phase may change with time as the nitrogen is replaced with air. Thus, C^* will no longer be constant.

END

Aeration & Agitation:

Determination of K_La values-

Gassing-out Techniques-2

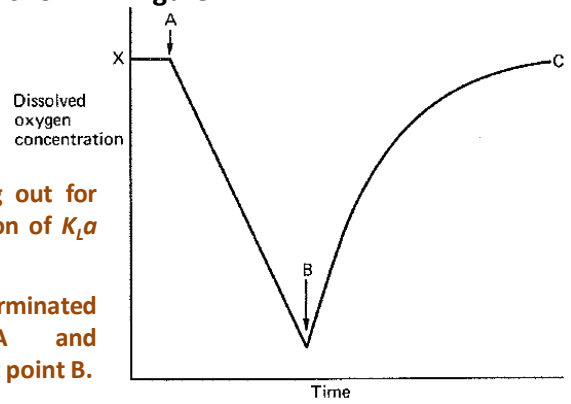
ii. Dynamic Method of Gassing-out

Respiratory activity of a growing culture in the fermenter is utilized to lower the oxygen level prior to aeration. Therefore, the estimation has the advantage of being carried out during a fermentation which should give a more realistic assessment of the fermenter's efficiency.

Because of the complex nature of fermentation broths the probe used to monitor the change in dissolved oxygen concentration must be of the membrane-covered type which may necessitate the use of the response-correction factors.

Aeration & Agitation

The procedure involves stopping the supply of air to the fermentation which results in a linear decline in the dissolved oxygen concentration due to the respiration of the culture, as shown in Figure:



Dynamic gassing out for the determination of K_La values.

Aeration was terminated at point A and recommenced at point B.

Aeration & Agitation

The slope of the line AB in above Figure a measure of the respiration rate of the culture. At point B the aeration is resumed and the dissolved oxygen concentration increases until it reaches concentration X. Over the period, BC, the observed increase in dissolved oxygen concentration is the difference between the transfer of oxygen into solution and the uptake of oxygen by the respiring culture as expressed by the equation:

$$dC_L/dt = K_La(C^* - C_L) - xQ_{O_2}$$

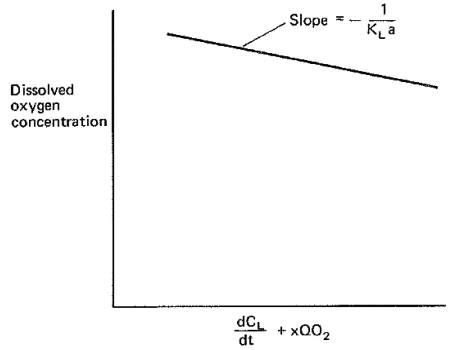
where x is the concentration of biomass and
Q_{O_2} is the specific respiration rate (mmoles of oxygen g^{-1} biomass h^{-1}).

Aeration & Agitation

The term xQ_{O_2} is given by the slope of the line AB in previous referred Figure. Equation in previous slide may be rearranged as:

$$C_L = -1/K_La\left\{(dC_L/dt) + xQ_{O_2}\right\} + C^*$$

Now, from this equation, a plot of C_L *versus* $(dC_L/dt) + xQ_{O_2}$ will yield a straight line, the slope of which will equal $-1/K_La$, as shown in Figure:

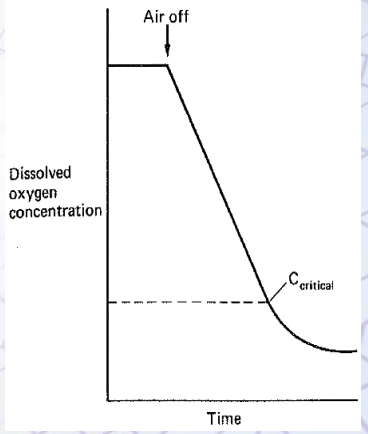


Aeration & Agitation

The dynamic gassing-out method has the advantage over the previous methods of determining the K_La during an actual fermentation and may be used to determine K_La values at different stages in the process. The technique is also rapid and only requires the use of a dissolved-oxygen probe, of the membrane type.

A major limitation in the operation of the technique is the range over which the increase in dissolved oxygen concentration may be measured. It may be difficult to apply the technique during a fermentation which has an oxygen demand close to the supply capacity of the fermenter.

Both the dynamic and static methods are also unsuitable for measuring K_La values in viscous systems.



The occurrence of oxygen limitation during the dynamic gassing out of a fermentation.

END

Aeration & Agitation:

Determination of K_La values-

The Oxygen Balance Technique

The K_La of a fermenter may be measured during a fermentation by the oxygen balance technique which determines, directly, the amount of oxygen transferred into solution in a set time interval. The procedure involves measuring the following parameters:

i. The volume of the broth contained in the vessel, V_L (dm^3).
ii. The volumetric air flow rates measured at the air inlet and outlet, Q_i and Q_o respectively ($dm^3\ min^{-1}$).
iii. The total pressure measured at the fermenter air inlet and outlet, P_i and P_o, respectively (atm. absolute).
iv. The temperature of the gases at the inlet and outlet, T_i and T_o, respectively (K).
v. The mole fraction of oxygen measured at the inlet and outlet, Y_i and Y_o respectively.

The oxygen transfer rate may then be determined from the following equation:

$$\mathrm{OTR} = \left(7.32 \times 10^5 / V_L\right)\left(Q_i P_i y_i / T_i - Q_o P_o y_o / T_o\right)$$

Where, 7.32 X 10^5 is the conversion factor equaling (60 min h^{-1}) [mole/22.4 dm^3 (STP)] (273 K/1 atm).

These measurements require accurate flow meters, pressure gauges and temperature-sensing devices as well as gaseous oxygen analyzers. The ideal gaseous oxygen analyzer is a mass spectrometer analyzer which is sufficiently accurate to detect changes of 1 to 2%. The K_La may be determined, provided that C_L and C^* are known, from equation $dC_L/dt = K_La(C^* - C_L)$:

$$\mathrm{OTR} = K_La(C^* - C_L) \text{ or } K_La = \mathrm{OTR}/(C^* - C_L)$$

Aeration & Agitation

The oxygen-balance technique appears to be the simplest method for the assessment of K_La and has the advantage of measuring aeration efficiency during a fermentation.

Aeration & Agitation

Factors affecting K_La the air-flow rate employed in vessels,

–the degree of agitation inside vessels,

– the rheological properties of the culture broth and

– the presence of antifoam agents

END

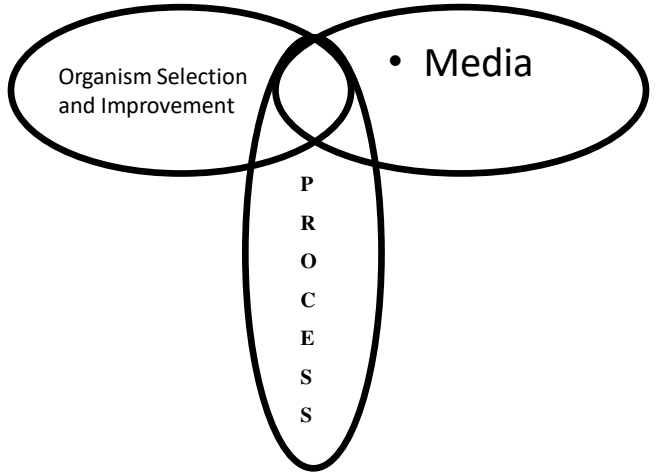
Advances in Fermentation Technology

Process Organism:

Selection criteria

Process Organism

Overview



Process Organism

Microorganisms are ubiquitous in nature. Thus, the diversity of micro-organisms may be exploited by searching for strains from the natural environment able to produce products of commercial value.

Process Organism

The selection of the culture to be used is a compromise between the productivity of the organism and the economic constraints of the process.

Process Organism

Bull *et al.* (1979) cited a criteria as being important in the choice of process organism:

1. The nutritional characteristics of the organism. It is frequently required that a process be carried out using a very cheap medium or a pre-determined one, e.g. the use of methanol as an energy source. These requirements may be met by the suitable design of the isolation medium.
2. The optimum temperature of the organism. The use of an organism having an optimum temperature above 40° considerably reduces the cooling costs of a large-scale fermentation and, therefore, the use of such a temperature in the isolation procedure may be beneficial.

Process Organism

3. The reaction of the organism with the equipment to be employed and the suitability of the organism to the type of process to be used.
4. The stability of the organism and its amenability to genetic manipulation.
5. The productivity of the organism, measured in its ability to convert substrate into product and to give a high yield of product per unit time.
6. The ease of product recovery from the culture.

Process Organism

The Question is?

How we get organism who meet the Bull *et al*., (1979) criteria.

Answer will be:

Isolate from the nature or collect from the stock cultures.

Process Organism

The first stage in the screening for microorganisms of potential industrial application is their **isolation**. Isolation involves obtaining either pure or mixed cultures followed by their assessment to determine which carry out the desired reaction or produce the desired product.

Process Organism

In some cases it is possible to design the isolation procedure in such a way that the growth of producers is encouraged or that they may be recognized at the isolation stage, whereas in other cases, organisms must be isolated and producers recognized at a subsequent stage. However, it should be remembered that the isolate must eventually carry out the process economically.

Process Organism

However, before the process may be put into commercial operation the **toxicity** and **product potential** of the organism has to be assessed.

END

Process Organism:

Collection
of
Industrially Important
Microorganisms-1

As pertained to the discussion in previous module, it implies that cultures must be isolated, in some way, from natural environments.
but
It can be collected from culture banks/stocks.

List of Some of the Major Culture Collections

Culture Collection	Address
National Collection of Type cultures (NCTC)	PHLS Central Public Health Laboratory, 61 Colindale Avenue, London NW9 5HT, UK
National Collections of Industrial and Marine Bacteria Ltd. (NCIB, NCMB)	23 St Machar Drive, Aberdeen AB2 1RY, UK
National Collection of Yeast Cultures (NCYC)	AFRC Institute of Food Research, Norwich Laboratory, Colney Lane, Norwich NR4 7UA, UK
American Type Culture Collection (ATCC)	12301 Parklawn Drive, Rockville, MD 20852, USA

List of Some of the Major Culture Collections

Culture Collection	Address
Collection of International Mycological Institute (IMI)	Culture Collection and Industrial Service Division, Ferry Lane, Kew, Surrey TW9 3AF, UK
Culture Collection of the Institute for Fermentation (IFO)	Institute for Fermentation, 17-85 Jugo-Honchmachi, 2-chome, Yodogawa-ku, Osaka, Japan
Agricultural Research Service Culture Collection (ARS/NRRL)	ARS Culture Collection National Center for Agricultural Utilization Research 1815 N. University Street Peoria, IL 61604, USA

Process Organism

Most of the Culture collection facilities as mentioned in last slide provide pure prokaryotic and eukaryotic microbial cultures free of cost for research studies only. Some of them might charge shipping costs accordingly.

Such collections may provide organisms of known characteristics but may not contain those possessing the most desirable features, whereas the environment contains a myriad of organisms, very few of which may be satisfactory. It is certainly cheaper to buy a culture than to isolate from nature, but it is also true that a superior organism may be found after an exhaustive search of a range of natural environments.

Process Organism

However, it is always worthwhile to purchase cultures demonstrating the desired characteristics, however weakly, as they may be used as model systems to develop culture and assay techniques which may then be applied to the assessment of natural isolates.

END

Advances in Fermentation Technology

Process Organism:

Collection
of
Industrially Important
Microorganisms-2

Process Organism

The ideal isolation procedure commences with an environmental source (frequently soil) which is highly probable to be rich in the desired types, is so designed as to favour the growth of those organisms possessing the industrially important characteristic (i.e. the industrially useful characteristic is used as a selective factor) and incorporates a simple test to distinguish the most desirable types.

Selective pressure may be used in the isolation of organisms which will grow on particular substrates, in the presence of certain compounds or under cultural conditions adverse to other types.

Process Organism

INDUSTRIAL STRATEGY FOR USAGE OF MICROORGANISMS

Isolation of microbes from natural source

↓

Identification of desired microbes

↓

Characterization of microbes

↓

Screening of desired microbes

↓

Strain Improvement

↓

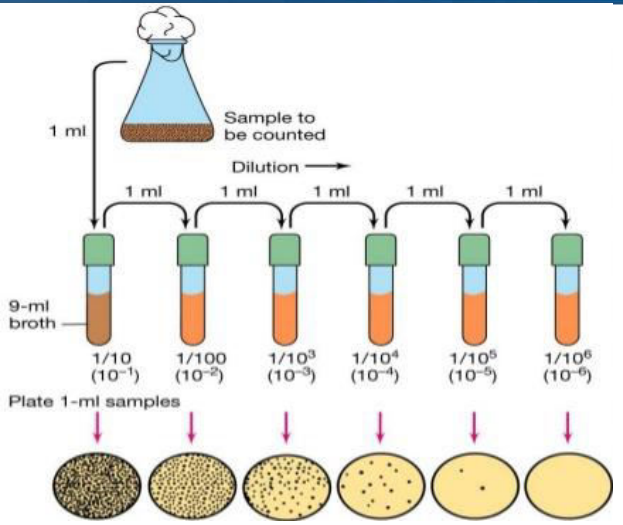
Fermentation

Process Organism

Key strategy to isolate a target microorganism from a specific environmental sample/site

Thermophiles	Hot spring, thermophilic vents compost
Alkalophiles	High salinity soils and wells
Acidophiles	Pine growing sites, fruits
Osmophiles	Seas and oceans
Yeast	Wine yards
Phosphate solubilizers	Agricultural soil, rhizosphere
Protease producers	Dairy industry, meat processing units.
Enzymes like cellulase, pectinase, xylanase etc.	Agricultural soil, paper and pulp industry, waste disposal area of fruit juice plants, grain market.

Process Organism



Process Organism

10-folds serial dilution of natural or enriched sample and its inoculation on selective medium is one of the most practically applicable procedure for initial isolation of desired or all types of microbes from different samples, which gives a mixed microbial population most of the times.

END

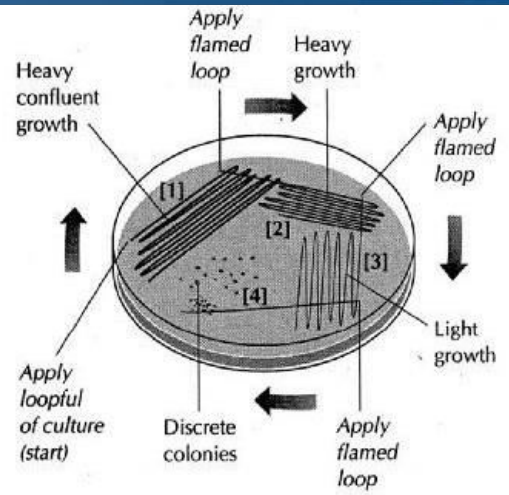
Process Organism:

collection
of
Industrially Important
Microorganisms-3

Process Organism

The isolation method most commonly used to get pure cultures is the streak plate method (Figure on next slide). A sterile inoculating loop is dipped into a mixed culture that contains more than one type of microbe and is streaked in a pattern over the surface of the nutrient medium. As the pattern is traced, bacteria are rubbed off the loop onto the medium. The last cells to be rubbed off the loop are far enough apart to grow into isolated colonies. These colonies can be picked up with an inoculating loop and transferred to a test tube of nutrient medium to form a pure culture containing only one type of bacterium.

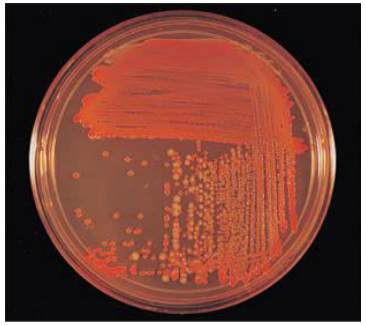
Process Organism



Streak plate method for pure-culturing of required microorganism is done for non-filamentous unicellular microorganisms only.

Process Organism





Process Organism

The most desirable isolation medium would be one which selects for the desired types and also allows maximum genetic expression. Cultures grown on such media could then be used directly in a screen.

Process Organism

However, it is more common that, once isolated, the organisms are grown on a range of media designed to enhance productivity.

END

Advances in Fermentation Technology

Process Organism:
Screening Methods for Isolated Microbes

Process Organism

Early screening strategies tended to be empirical, labour intensive and had relatively low success rates. As the number of commercially important compounds isolated increased, the success rates of such screens decreased further.

Thus, new screening methods have been developed which are more precisely targeted to identify the desired activity. The evolution of antibiotic screens serves as an excellent illustration of the development of more precise, targeted systems.

Process Organism

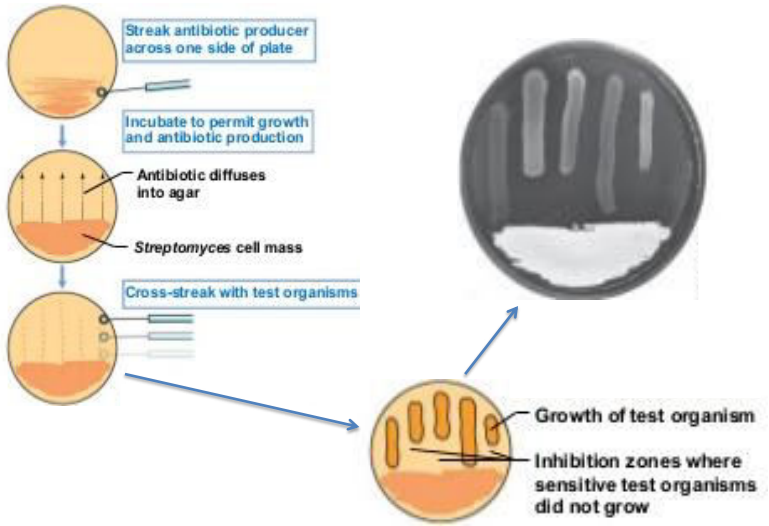
Antibiotics were initially detected by growing the potential producer on an agar plate in the presence of an organism (or organisms) against which antimicrobial action was required.

Production of the antibiotic was detected by inhibition of the test organism(s).

Alternatively, the microbial isolate could be grown in liquid culture and the cell-free broth tested for activity.

This approach was extended by using a range of organisms to detect antibiotics with a defined antibacterial spectrum.

Process Organism



Process Organism

The progress in molecular biology, genetics and immunology has also contributed extensively to the development of innovative screens, by enabling the construction of specific detector strains, increasing the availability of enzymes and receptors and constructing extremely sensitive assays.

Process Organism

Bull (1992) summarized the major contributions as follows:

1. The provision of test organisms that have increased sensitivities, or resistances, to known agents. For example, the use of super-sensitive strains for the detection of β-lactam antibiotics.
2. The cloning of genes coding for enzymes or receptors that may be used in inhibitor or binding screens makes such materials more accessible and available in much larger amounts.
3. The development of reporter gene assays. A reporter gene is one which codes for an easily assayable product so that it can be used to detect the activation of a control sequence to which it is fused.

Process Organism

4. Molecular probes for particular gene sequences may enable the detection of organisms capable of producing certain product groups.
5. The development of immunologically based assays such as ELISA.

END

Advances in Fermentation Technology

Process Organism:
Identification of Microorganisms

Process Organism

Various techniques are employed to characterize and identify microorganisms:

- Phenotypic characteristics
  - Microscopic morphology
  - Metabolic differences (Biochemical Testing)
  - Serology
  - Fatty acid analysis
- Genotypic characteristics
  - Nucleic acid probes
  - DNA amplification
  - Ribotyping

Process Organism

Genotypic Characterization (Nucleic acid probes)

- Used to locate unique sequences
  - Single-stranded DNA (or RNA)
- Generally labeled (radioactive or fluorescent)
  - Complementary to the sequence of interest
  - Observe and identify intact microorganisms
- Fluorescence in situ hybridization (FISH)
  - Observe and identify samples
- DNA amplification by PCR
  - Specific primers are used

Process Organism

- Sequencing rRNA genes (16S rRNA for bacteria)
  - Evolutionarily highly conserved genes Variable regions are used to identify an organism. Particularly useful in identifying microbes that are difficult to culture.

END

Advances in Fermentation Technology

Process Organism:

Storage/Preservation of Microbial Cultures

Process Organism

- The isolation of a suitable organism for a commercial process may be a long and very expensive procedure and it is therefore essential that it retains the desirable characteristics that led to its selection.

- Also, the culture used to initiate an industrial fermentation must be viable and free from contamination. Thus, industrial cultures must be stored in such way as to eliminate genetic change, protect against contamination and retain viability.

Process Organism

- An organism may be kept viable by repeated sub-culture into fresh medium, but, at each cell division, there is a small probability of mutations occurring and because repeated sub-culture involves very many such divisions, there is a high probability that strain degeneration would occur.

- Also repeated sub-culture carries with it the risk of contamination. Thus, preservation techniques have been developed to maintain cultures in a state of 'suspended animation' by storing either at reduced temperature or in a dehydrated form.

Process Organism

❑ Storage at Reduced Temperature

▪Storage on Agar Slants

Cultures grown on agar slopes may be stored in a refrigerator (5°C) or a freezer (- 20°C) and sub-cultured at approximately 6-monthly intervals. The time of subculture may be extended to 0I year if the slopes are covered with sterile medicinal grade mineral oil.

▪Storage under Liquid Nitrogen

The metabolic activities of micro-organisms may be reduced considerably by storage at the very low temperatures (-150° to -196°) which may be achieved using a liquid nitrogen refrigerator.

Process Organism

Fungi, bacteriophage, viruses, algae, yeasts, animal and plant cells and tissue cultures can all be successfully preserved by using this method. The technique involves growing a culture to the maximum stationary phase, resuspending the cells in a cryoprotective agent (such as 10% glycerol) and freezing the suspension in sealed ampoules before storage under liquid nitrogen.

Process Organism

❑ Storage in a Dehydrated Form

▪Dried Cultures

Dried soil cultures have been used widely for culture preservation, particularly for sporulating mycelial organisms. Moist, sterile soil may be inoculated with a culture and incubated for several days for some growth to occur and then allowed to dry at room temperature for approximately 2 weeks. The dry soil may be stored in a dry atmosphere or, preferably, in a refrigerator. The technique has been used extensively for the storage of fungi and actinomycetes and Pridham *et al.* (1973) observed that of 1800 actinomycetes dried on soil about 50% were viable after 20-years storage.

Process Organism

❑ Storage in a Dehydrated Form

▪Lyophilization

During lyophilization (freeze-drying), a suspension of microbes is quickly frozen at temperatures ranging from -54°C to -72°C, and the water is removed by a high vacuum (sublimation). While under vacuum, the container is sealed by melting the glass with a high-temperature torch. The remaining powder like residue that contains the surviving microbes can be stored for years. The organisms can be revived at any time by hydration with a suitable liquid nutrient medium.

Process Organism

Lyophilization is very convenient for service culture collections because, once dried, the cultures need no further attention and the storage equipment (a refrigerator) is cheap and reliable.

END

Advances in Fermentation Technology

Process Organism:

Improvement of Industrial Important Microorganisms-1

Process Organism

Natural isolates usually produce commercially important products in very low concentrations and therefore every attempt is made to increase the productivity of the chosen organism. Increased yields may be achieved by optimizing the culture medium and growth conditions, but this approach will be limited by the organism's maximum ability to synthesize the product.

The potential productivity of the organism is controlled by its genome and, therefore, the genome must be modified to increase the potential yield. The cultural requirements of the modified organism would then be examined to provide conditions that would fully exploit the increased potential of the culture, while further attempts are made to beneficially change the genome of the already improved strain.

Process Organism

Thus, the process of strain improvement involves the continual genetic modification of the culture, followed by reappraisals of its cultural requirements.

Process Organism

Improvement of strains can be put down in simple term as follows:

- regulating the activity of the enzymes secreted by the organisms;
- in the case of metabolites secreted extra-cellularly, increasing the permeability of the organism so that the microbial products can find these way more easily outside the cell;
- selecting suitable producing strains from a natural population;
- manipulation of the existing genetic apparatus in a producing organism;
- introducing new genetic properties into the organism by recombinant DNA technology or genetic engineering.

Process Organism

Methods of manipulating the genetic apparatus of industrial organisms

A. Methods not involving foreign DNA
   1. Conventional mutation/Random Mutagenesis

B. Methods involving DNA foreign to the organism (i.e. recombination)
   2. Transduction
   3. Conjugation
   4. Transformation
   6. Protoplast fusion
   7. Genetic engineering
   8. Metabolic engineering
   9. Site-directed mutagenesis

Process Organism

Detail of some of the above mentioned methods will be discussed in next modules.

END

Advances in Fermentation Technology

Process Organism:
Improvement of Industrial Important Microorganisms-2

Process Organism

❑ Mutation and selection

- Mutations; can occur spontaneously or can be induced by chemical and physical agents involves change in the genetic material, might cause reduction, enhancement or loss of gene activity.
- In order to isolate mutants selection system is required.
- Nutritionally defective, resistant, temperature sensitive and similar types of mutations that are used in basic research are relatively simple.

The establishment of selection system for mutations which aim to improve the yield in the production of given primary and secondary metabolites or certain enzymes is more complicated.

Process Organism

Chemical mutagens (alkylating agents, base analogues, deaminating agents) or Physical mutagens (UV and the Ionizing radiations: X-rays, gamma rays, alpha-particles and fast neutrons) could be used for increasing the mutation rate of bacteria. Then we can select the ones with desired properties.

Process Organism

❑ Choice of a Mutagen

Mutagenic agents are numerous but not necessarily equally effective in all organisms. Other factors besides effectiveness to be borne in mind are:

a) the safety of the mutagen: many mutagens are carcinogens, b) simplicity of technique, and c) ready availability of the necessary equipment and chemicals.

Among physical agents, UV is to be preferred since it does not require much equipment, and is relatively effective and has been widely used in industry. The disadvantage of UV is that it is absorbed by glass; it is also not effective in opaque or colored organisms. Chemical methods other than NTG (nitrosoguanidine) are probably best used in combination with UV.

Process Organism

❑ Random Mutagenesis

Early approaches to mutagenesis rely on methods which are entirely random in the mutations produced. Cells or organisms may be exposed to mutagens such as UV radiation or mutagenic chemicals, and mutants with desired characteristics are then selected. For *Escherichia coli*, mutants may be selected first by exposure to UV radiation, then plated onto agar medium. The colonies formed are then replica-plated, one in rich medium, another in minimal medium, and mutants that have specific nutritional requirements can then be identified by their inability to grow in minimal medium. Similar procedures may be repeated with other types of cells and with different media for selection.

Process Organism

Alkylating agents such as *N*-ethyl-*N* nitrosourea (ENU) have been used to generate mutant mice. Ethyl methanesulfonate (EMS) is also often used to generate animal and plant mutants.

END

Advances in Fermentation Technology

Process Organism:
Improvement of Industrially Important Microorganisms-3

Process Organism

❑ Site-Directed Mutagenesis

Many researchers seek to introduce selected changes to DNA in a site-specific manner. Analogs of nucleotides and other chemicals were first used to generate localized point mutations. Such chemicals include aminopurine, which induces AT to GC transition, while nitrosoguanidine, bisulfite, and N^4-hydroxycytidine may induce GC to AT transition.

These techniques allow specific mutations to be engineered into a protein; however, they are not flexible with respect to the kinds of mutants generated, nor are they as specific as later methods of site-directed mutagenesis and therefore have some degree of randomness.

Process Organism

- Current techniques for site-specific mutation commonly involve using mutagenic oligonucleotides in a primer extension reaction with DNA polymerase.
- This methods allows for point mutation, or deletion or insertion of small stretches of DNA to be introduced at specific sites.
- Advances in methodology have made such mutagenesis now a relatively simple and efficient process.

Process Organism

❑ **Protoplast Fusion**

Protoplasts are formed from bacteria, fungi, yeast and actinomycetes when dividing cells are caused to lose their cell walls. Fusion from mixed populations of protoplasts is greatly enhanced by the use of Polyethyleneglycol (PEG). **The method has great industrial potential and has experimentally been used to achieve higher yields of antibiotics through fusion with protoplasts from different fungi. Protoplast fusion has been demonstrated as an efficient way to induce hetero-karyon formation and recombination with high frequency.**

Process Organism

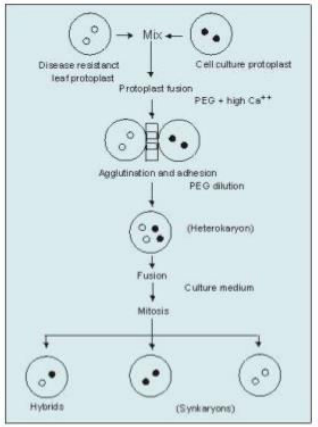
Protoplast fusion has been successfully done with *Bacillus subtilis* and *Bacillus megaterium*; among several species of *Streptomyces spp*. like *S. coeli-color*, *S. acrimycini*, *S. olividans*, *S. pravulies*; between the fungi *Geotrichum* and *Aspergilus*; and Yeasts.

Kim *et al.* (1998) did a comparative study on strain improvement of *Aspergillus oryzae* for protease production by both mutation and protoplast fusion.

- **UV radiation = 14 times higher yield**
- **EMS = 39 times higher yield**
- **Protoplast fusion using PEG and $CaCl_2$ = 82 times higher yield**

Process Organism

- Bacteria-grown plasmids are introduced into animal cells via fusion with bacterial protoplasts
- Bacterial cell walls are removed using lysozyme treatment
- Protoplasts fuse to the mammalian cells
- DNA purification not required
- may introduce bacterial components into recipient cells, cultures may be overgrown with bacteria
- *Bacteria are used to produce plasmids containing the human gene of interest. The cell walls are removed using lysozyme, resulting in the formation of protoplasts. The protoplasts are brought in contact with the mammalian cells, leading to protoplast fusion to mammalian cells, using Polyethylene glycol (PEG).*



Process Organism

Auxotrophy is the inability of an organism to synthesize a particular organic compound required for its growth. An auxotroph is an organism that displays this characteristic.

Process Organism

Prototrophy is characterized by the ability to synthesize all the compounds needed for growth. Protrophic organism is a wild-type organism expressing its natural phenotype.

END

Advances in Fermentation Technology

Process Organism:

Mutant Selection-1

Process Organism

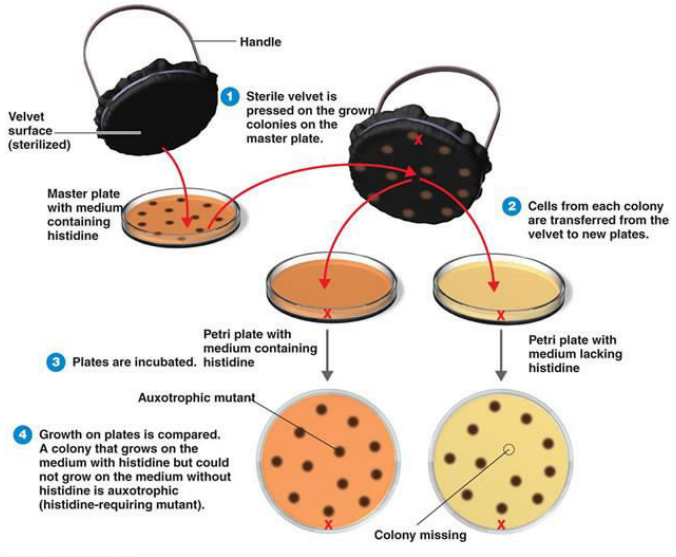
Following exposure to the mutagen the cells should be suitably diluted and plated out to yield 50 – 100 colonies per plate. The selection of mutants is greatly facilitated by relying on the morphology of the mutants or on some selectivity in the medium. When morphological mutants are selected, it is in the hope that the desired mutation is pleotropic (i.e., a mutation in which change in one property is linked with a mutation in another character). The classic example of a pleotropic mutation is to be seen in the development of penicillin-yielding strains of *Penicillium chrysogenum*. It was found in the early days of the development work on penicillin production that after irradiation, strains of *Penicillium chrysogenum* with smaller colonies and which also sporulated poorly were better producers of penicillin.

Process Organism

❑ Isolation/Selection of Auxotrophic Mutants by Replica Plating Method

Mutants can be detected by selecting or testing for an altered phenotype. Positive selection involves the selection of mutant cells and rejection of non-mutated cells. To identify mutants that have gained some physical trait, Replica plating is used for negative selection. Negative selection identifies mutants that have lost some physical trait - for example, auxotrophs that have nutritional requirements not possessed by the parent (nonmutated) cell.

Process Organism



Process Organism

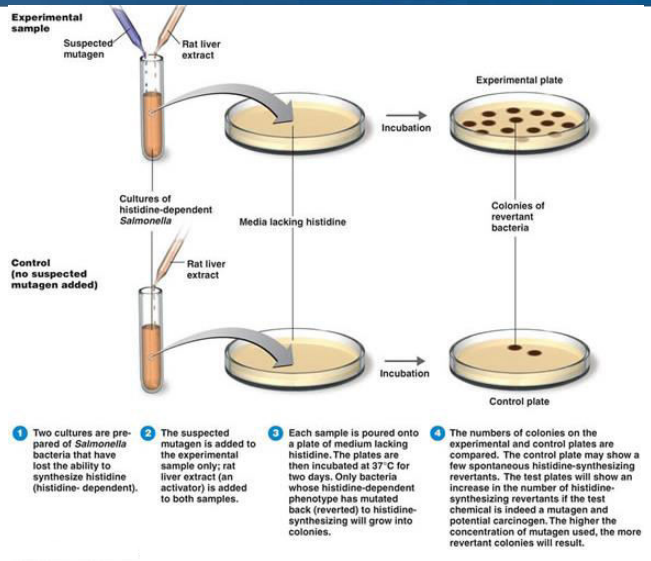
❑ Ames Test

The Ames test is a relatively inexpensive and rapid test for identifying possible chemical carcinogens.

The test assumes that a mutant cell can revert to a normal cell in the presence of a mutagen and that many mutagens are carcinogens.

Histidine auxotrophs (require histidine in the media) of *Salmonella* are exposed to an enzymatically treated potential carcinogen, and reversions to the nonmutant state are selected.

Process Organism



Process Organism

Before considering the methods used for the selection of methods producing improved levels of primary metabolites it is necessary to study the mechanisms of control of their biosynthesis such that the 'blueprints', may be drawn accurately.

END

Process Organism:

Development of Inocula for Industrial Fermentations Introduction 1

Development of Inocula

It is essential that the culture used to inoculate a fermentation satisfies the following criteria:

1. It must be in a healthy, active state thus minimizing the length of the lag phase in the subsequent fermentation.
2. It must be available in sufficiently large volumes to provide an inoculum of optimum size.
3. It must be in a suitable morphological form.
4. It must be free of contamination.
5. It must retain its product-forming capabilities.

Development of Inocula

The quantity of inoculum normally used is between 3 and 10% of the medium volume. A relatively large inoculum volume is used to minimize the length of the lag phase and to generate the maximum biomass in the production fermenter in as short a time as possible, thus increasing vessel productivity. Thus, starting from a stock culture, the inoculum must be built up in a number of stages to produce sufficient biomass to inoculate the production-stage fermenter. This may involve two or three stages in shake flasks and one to three stages in fermenters, depending on the size of the ultimate vessel. Throughout this procedure there is a risk of contamination and strain degeneration necessitating stringent quality-control procedures.

Development of Inocula

The greater the number of stages between the master culture and the production fermenter the greater the risk of contamination and strain degenration. Therefore, a compromise may be reached regarding the size of the inoculum to be used and the risk of contamination and strain degeneration.

Another factor to be considered in the determination of the inoculum volume is the economics of the process. A seed fermenter 10% of the size of the production fermenter represents a considerable financial investment and must be justified in terms of productivity.

Development of Inocula

A large-scale continuous fermentation for the production of biomass would be expected to operate at steady state in excess of 100 days.

Thus, the number of times that the fermenter inoculated should be very few compared with batch or fed-batch systems. In such circumstances it may be more economic to compromise on the size of the inoculum and to tolerate a relatively lengthy period of growth up to maximum biomass than to invest a large seed vessel which would be used on very few occasions.

Development of Inocula

This is particularly relevant for biomass continuous processes where one very large fermenter may be used and, thus, any seed vessel would only be servicing the one production vessel.

END

Advances in Fermentation Technology

Development of Inocula

FOR INDUSTRIAL FERMENTATIONS-
Introduction-2

Development of Inocula

❑Description of Typical inoculum-development

The master culture is reconstituted and plated on to solid medium; approximately ten colonies of typical morphology of high producers are selected and inoculated on to slopes as the sub-master cultures, each sub-master culture being used for a new production run.

At this stage, shake flasks may be inoculated to check the productivity of these cultures, the results of such tests being known before the developing inoculum eventually reaches the production plant. A sub-master culture is used to inoculate a shake flask (250 or 500 cm^3 containing 50 or 100 cm^3 medium) which, in turn, is used as inoculum for a larger flask, or a laboratory fermenter, which may then be used to inoculate a pilot-scale fermenter.

Development of Inocula

Culture purity checks are carried out at each stage to detect contamination as early as possible. Although the results of these tests may not be available before the culture has reached the production plant, at least it is known at which stage in the procedure contamination occurred. For a sporulating organism the process may be modified to facilitate the use of a spore suspension as inoculum.

Lincoln (1960) suggested a more elaborate procedure for the development of inoculum for bacterial fermentations which, with minor modifications, is applicable to any type of culture.

Development of Inocula

The procedure involved the use of one sub-master culture to develop a bulk inoculum which was subdivided, stored in a frozen state and used as inocula for several months. A single colony, derived from a sub-master culture, was inoculated into liquid medium and grown to maximum log phase. This culture was then transferred into nineteen times its volume of medium and incubated again to the maximum log phase, at which point it was dispensed in 20 cm^3 volumes, plug frozen and stored at below –20°C. At least 3% of the samples were tested for purity and productivity in subsequent fermentation and, provided these were suitable, the remaining samples could be used as initial inocula for subsequent fermentations.

Development of Inocula

To use one of the stored samples as inoculum it was thawed and used as a 5% inoculum for a seed culture which, in turn, was used as a 5% inoculum for the next stage in the programme.

END

Advances in Fermentation Technology

Development of Inocula

FOR INDUSTRIAL FERMENTATIONS-

Criteria for the Transfer of Inoculum

Development of Inocula

The physiological condition of the inoculum when it is transferred to the next culture stage can have a major effect on the performance of the fermentation. The optimum time of transfer must be determined experimentally and then procedures established so that inoculation with an ideal culture may be achieved routinely.

These procedures include the standardization of cultural conditions and monitoring the state of an inoculum culture so that it is transferred at the optimum time, i.e. in the correct physiological state. The most widely used criterion for the transfer of vegetative inocula is biomass and such parameters as packed cell volume, dry weight, wet weight, turbidity, respiration, residual nutrient concentration and morphological form have been used.

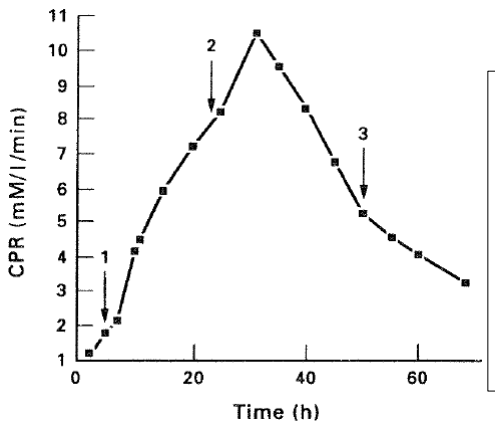
Development of Inocula

Criteria which may be monitored on-line are the most convenient parameters to use as indicators of inoculum quality and these would include:

- dissolved oxygen, pH (although pH would normally be controlled in seed fermentations) and oxygen or carbon dioxide in the effluent gas.

Parton and Willis (1990) advocated the use of the carbon dioxide production rate (CPR) as a transfer criterion which requires analysis of the fermenter effluent air. This approach is suitable only when transfer is being made from a fermenter. These workers provide an excellent example of the effect of inoculum transfer time on the productivity of a streptomycete secondary metabolite, as shown in Figures in next slides.

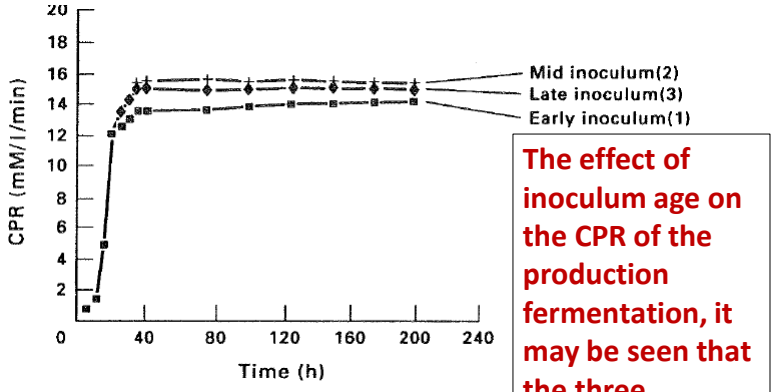
Development of Inocula



The carbon dioxide production rate (CPR) profile of the inoculum culture showing the points 0, 2 and 3) at which inocula were removed.

The effect of inoculum age on growth and productivity in a streptomycete fermentation.

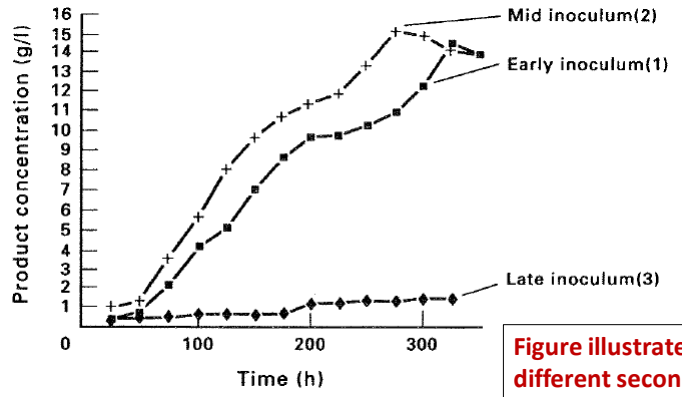
Development of Inocula



The effect of inoculum age on the CPR of the production fermentation, it may be seen that the three fermentations performed similarly.

The effect of inoculum age on growth and productivity in a streptomycete fermentation.

Development of Inocula



The effect of inoculum age on growth and productivity in a streptomycete fermentation.

Figure illustrates very different secondary metabolite production of the three fermentations.

Development of Inocula

Thus, although the time of transfer had only a marginal influence on biomass in the production fermentation the effect on product formation was critical.

END

Advances in Fermentation Technology

Development of Inocula

Examples

Development of Inocula

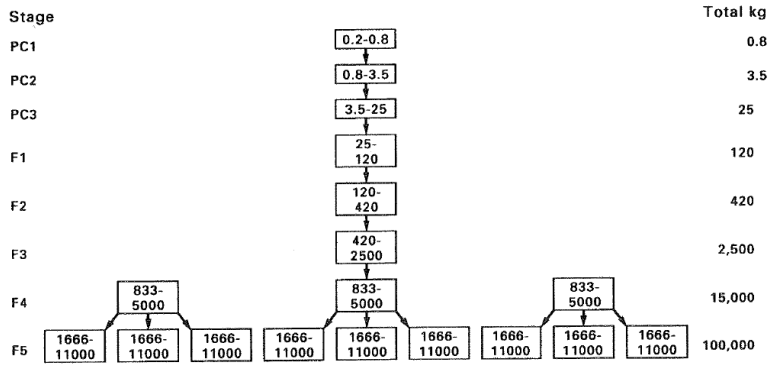
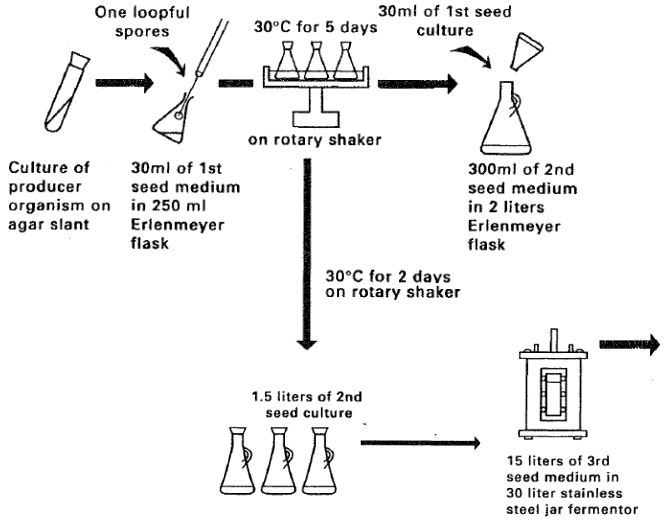
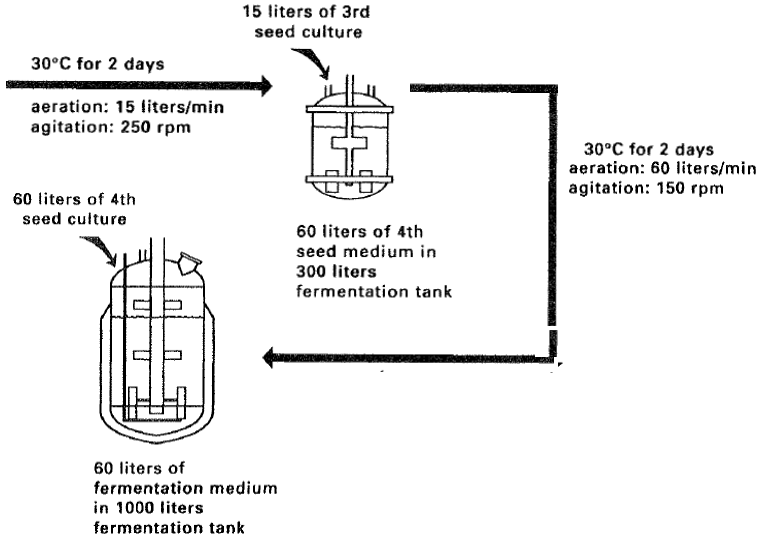


Figure showing the development of inoculum for the commercial production of bakers' yeast. PC 1, 2 and 3 are pure culture batch fermentations. F1 and 2 are non-aseptic batch fermentations. F3 and 4 are fed-batch fermentations and F5 is the final fed-batch fermentation

Development of Inocula



Development of Inocula



Development of Inocula

The choice of inoculum for the production stage depends on various factors such as the length of the cycle of the fermentation process, plant size and the availability and cost of labour.

END